

An Exact Computer-Assisted Proof of the Dittert Conjecture for $n = 4$

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Review status. This document reports a new exact computational certificate produced in an interactive research session. The certificate and three exact verifiers are included, two of which independently expand the saved factors as literal squares. The argument has not yet undergone independent peer review; it should be checked before being cited as an established result.

Abstract

For a nonnegative 4×4 matrix A whose entries sum to 4, let

$$\phi(A) = \prod_{i=1}^4 r_i + \prod_{j=1}^4 c_j - \text{per}(A),$$

where r_i and c_j are the row and column sums. We give an exact computer-assisted proof that

$$\phi(A) \leq 2 - \frac{4!}{4^4} = \frac{61}{32},$$

with equality only at the matrix J_4 whose entries are $1/4$. Published structural results reduce the boundary case to a single face with two zero entries in distinct rows and columns. On that face, an integer sum-of-squares/copositive certificate proves a strict quartic inequality. The final verification uses arbitrary-precision integer arithmetic only; the smallest residual coefficient is

$$\frac{72\,694\,203\,872}{2^{60}} > 6.28 \times 10^{-8}.$$

1 Statement

Let

$$K_4 = \left\{ A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{4 \times 4} : \sum_{i,j} a_{ij} = 4 \right\}.$$

Write $J_4 = (1/4)_{i,j}$. The four-dimensional Dittert conjecture is the following statement.

Theorem 1. *For every $A \in K_4$,*

$$\phi(A) \leq \frac{61}{32}.$$

Equality holds if and only if $A = J_4$.

The proof combines two published structural facts with one exact polynomial certificate. Hwang proved that the only possible *positive* ϕ -maximizer is J_n [1]. Cheon and Wanless proved that partly decomposable matrices cannot maximize ϕ , and that a maximizer cannot have its complete zero set equal, up to row and column permutations, to one proper rectangular block [2, Theorems 3.2 and 3.7].

2 Reduction to one two-zero face

Lemma 2. *Let Z be a nonempty set of positions in a matrix. If Z contains no two positions lying in distinct rows and distinct columns, then all positions of Z lie in one row or all lie in one column. In particular, Z is one rectangular block.*

Proof. Choose $(i, j) \in Z$. Every other position in Z must share row i or column j with (i, j) . If both a position (i, j') with $j' \neq j$ and a position (i', j) with $i' \neq i$ existed, those two positions would lie in distinct rows and distinct columns. Hence only one of the two types can occur, and all of Z lies in row i or in column j . $\square$

Proposition 3. *To prove Theorem 1, it is enough to prove*

$$\phi(A) < \frac{61}{32} \quad \text{whenever} \quad A \in K_4, \quad a_{11} = a_{22} = 0. \quad (1)$$

Proof. The compactness of K_4 and continuity of ϕ give a global maximizer A . Since $J_4 \in K_4$, its maximum value is at least $\phi(J_4) = 61/32$.

If A is positive, Hwang's positive-support theorem gives $A = J_4$. Suppose instead that A is on the boundary. If its zero set contained no two positions in distinct rows and columns, Lemma 2 would put all zeroes in one row or one column. If that whole row or column were zero, the matrix would be partly decomposable; otherwise its complete zero set would be one proper rectangular block. Both alternatives contradict Cheon–Wanless. Thus it contains two zeroes in distinct rows and columns. Independent row and column permutations put those zeroes at $(1, 1)$ and $(2, 2)$ without changing ϕ . Inequality (1) would then give $\phi(A) < 61/32$, contradicting maximality. Therefore the maximizer is positive and hence equals J_4 . $\square$

3 Quartic formulation of the face

Let $X = A/4$, so the entries of X sum to 1. On the face $x_{11} = x_{22} = 0$, index the remaining fourteen entries by

$$V = \{(i, j) : 1 \leq i, j \leq 4\} \setminus \{(1, 1), (2, 2)\}.$$

Write x_v for $v \in V$ and $S = \sum_{v \in V} x_v$. Let $\mathcal{E}$ be the collection of four-element subsets $E \subseteq V$ that meet all four rows or meet all four columns. There are exactly 274 such sets. Define

$$F(x) = \sum_{E \in \mathcal{E}} \prod_{v \in E} x_v. \quad (2)$$

Lemma 4. *On this face,*

$$\prod_{i=1}^4 R_i + \prod_{j=1}^4 C_j - \text{per}(X) = F(x),$$

where R_i and C_j are the row and column sums of X . Consequently,

$$\phi(A) = 256F(x). \quad (3)$$

Proof. Expanding the product of row sums gives every square-free quartic choosing one cell from each row. Expanding the product of column sums gives every square-free quartic choosing one cell from each column. A monomial appearing in both expansions chooses one cell from every row and every column, so it is a permanent term and is subtracted once. Therefore each monomial in the union appears exactly once, proving (2). Scaling every entry by 4 multiplies both fourfold products and the permanent by $4^4 = 256$. $\square$

By (3), the required strict inequality is equivalent to

$$G(x) := \frac{61}{8192}S^4 - F(x) > 0 \quad \text{for } x \geq 0, S = 1. \quad (4)$$

4 The exact certificate

Let $m_2(x)$ denote the vector of all 105 degree-two monomials in the fourteen variables. The attached certificate contains:

- a group Γ of 16 permutations of the variables preserving the two-zero face and F ;
- one representative from each of the 16 orbits of unordered variable pairs under Γ ;
- an integer lower-triangular matrix $L_0 \in \mathbb{Z}^{105 \times 105}$;
- integer lower-triangular matrices $L_p \in \mathbb{Z}^{14 \times 14}$ for the 16 pair representatives.

All actual Cholesky factors are $2^{-30}L$. The integer-only verifier expands the following identity coefficient by coefficient:

$$\begin{aligned} 2^{60}G(x) &= \sum_{\gamma \in \Gamma} \left\| L_0^\top m_2(\gamma x) \right\|_2^2 \\ &\quad + \sum_{p=(u,v)} \sum_{\gamma \in \Gamma} x_{\gamma u} x_{\gamma v} \left\| L_p^\top(\gamma x) \right\|_2^2 + \sum_{|\alpha|=4} r_\alpha x^\alpha. \end{aligned} \quad (5)$$

Here every r_α is an integer and

$$\min_{|\alpha|=4} r_\alpha = 72\,694\,203\,872 > 0. \quad (6)$$

There are

$$\binom{14 + 4 - 1}{4} = 2380$$

quartic monomials, and the verifier checks every one.

Proposition 5. *The strict inequality (4) holds on the nonnegative simplex $S = 1$.*

Proof. Identity (5) is an exact polynomial identity over the integers. Its first sum is a sum of squares. Every summand in the second sum is nonnegative when $x \geq 0$, because it is a product of two nonnegative coordinates and a square norm. The final polynomial has strictly positive coefficients by (6). Since $S = 1$, at least one coordinate is positive, and the corresponding fourth-power monomial makes the final polynomial strictly positive. Hence $G(x) > 0$. $\square$

Proposition 5, Lemma 4, and Proposition 3 prove Theorem 1.

5 Exact verification and reproducibility

The numerical semidefinite search was used only to *discover* the factors. It is not part of the proof. The factors were rounded to integers, after which all proof decisions were made with arbitrary-precision integer arithmetic.

Three exact verification programs are supplied:

1. The primary verifier forms each exact Gram matrix $LL^\top$, expands the invariant polynomial, validates the symmetry group and all 274 hyperedges, and checks all 2380 residual coefficients.

2. The NumPy literal-square audit never forms a Gram matrix. It expands every saved factor literally and constructs the target directly from S^4 , the row and column products, and all 24 permanent permutations.
3. The third audit repeats the calculation without NumPy and parses the certificate archive itself.

All three return the same minimum coefficient in (6) and terminate with **CERTIFIED**. The certificate SHA-256 digest is

d76533bd1c5566ea8d96aa3b58a0b6a8bf3310eb438776ebd99a6bd88d5a11f6.

Run the checks from the bundle directory, without Python's `-O` option:

```
python verify_primary.py dittert_n4_exact_certificate.npz
python verify_literal_square_numpy.py dittert_n4_exact_certificate.npz
python verify_literal_square_stdlib.py dittert_n4_exact_certificate.npz
```

6 Scope and literature context

This certificate settles only the case $n = 4$. Separate July 2026 working material distributed with this note treats $n = 14, 15, 16$; Pang's June 2026 preprint treats all $n \geq 17$ [4], and Kafidov independently treats $n = 16$ [5]. No claim about dimensions 5 through 13 is made here. The published 2024 paper of Udayan and Somasundaram proves that a maximizing matrix in K_4 is fully indecomposable, but that stronger structural result is not needed here [3]. The present reduction uses only Hwang's positive-support theorem and Cheon–Wanless's two boundary exclusions.

References

- [1] S.-G. Hwang, *A note on a conjecture on permanents*, Linear Algebra and its Applications 76 (1986), 31–44.
- [2] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra and its Applications 436 (2012), 791–801.
- [3] D. K. Udayan and K. Somasundaram, *Lih Wang's and Dittert's conjectures on permanents*, Special Matrices 12 (2024), article 20240006.
- [4] Z. Pang, *Proof of Dittert's conjecture for dimensions $n \geq 17$* , arXiv:2606.01531 (2026), preprint.
- [5] B. Kafidov, *Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma*, arXiv:2607.19439 (2026), preprint.